Annual Review for Utility Rate Plan Exchange (URPX)

LF Energy TAC, 2026-10-13. Klaartje De Schepper, chair, URPX Working Group

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Agenda

- URPX: the project and its lifecycle position
- Incubation criteria: a self-assessment
- The past year: contributions and organizations
- How URPX is built and published: repositories, workflow, the ontology
- Key achievements, and where we could use help
- Feedback on working with LF Energy

Utility Rate Plan Exchange (URPX)

Description

Utility Rate Plan Exchange (URPX) aims to develop a comprehensive, standardized method for representing and exchanging utility rate plan data in machine-accessible format.

URPX will provide a solid yet versatile foundation that governments, utilities, vendors, and technology companies can leverage to improve transparency in energy pricing and clean energy procurement.

The project builds upon an existing collaboration between Flux Tailor and the National Laboratory of the Rockies (NLR) and aims to address the fragmented approach currently used by vendors who handle rate plan calculations in silos using custom logic and rules-based systems.

LFESS working group, accepted 2025-11-05. Working Group Draft since 2026-07-17.

Links

Website: https://urpx.org/

Standard: https://github.com/urpx-org/urpx

Working group: https://github.com/urpx-org/urpx-wg

Mailing list: https://lists.lfenergy.org/g/urpx-wg

LF Energy page: https://lfenergy.org/projects/utility-rate-plan-exchange-urpx/

Meetings: https://zoom-lfx.platform.linuxfoundation.org/meetings/utility-rate-plan-exchange

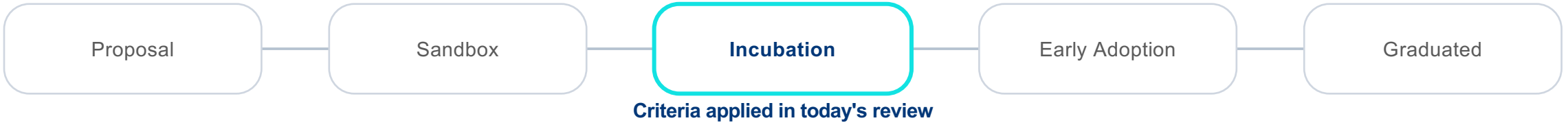
Artwork: https://artwork.lfenergy.org/projects/utility-rate-plan-exchange

Lifecycle position

LFESS Working Group status



LF Energy project lifecycle (stages defined for contributed projects)



URPX is registered as an LFESS Working Group; the five stages above are defined for contributed projects. **How a Working Group maps to the project stages is the open question for the TAC.**

Incubation Project review criteria

Self-assessment against the Incubation criteria. URPX is registered as an LFESS working group; the TAC lifecycle stages are defined for projects.

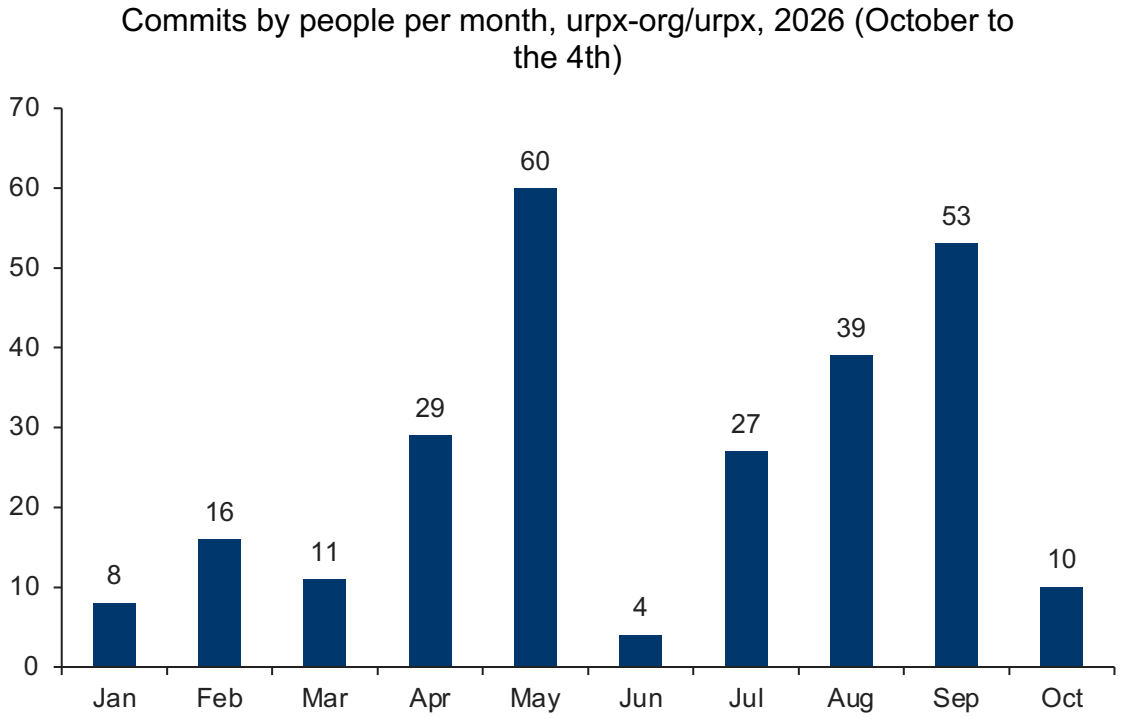
- **OpenSSF Best Practices badge: Passing.** Awarded at the public release before the meeting; the entry is filed on bestpractices.dev.
- **README in each repository: met.** README.md in urpx-org/urpx and urpx-org/urpx-wg.
- **GOVERNANCE file: met.** GOVERNANCE.md: chair, maintainers, voting committee, decision making; meetings open to anyone.
- **TSC members in LFX Project Control Center: not met.** The working group has a chair and a voting committee, not a TSC; no TSC committee is set up in LFX.
- **CODEOWNERS or COMMITTERS file: met.** .github/CODEOWNERS and MAINTAINERS.md in urpx-org/urpx; one maintainer today.

Incubation Project review criteria (continued)

- **Architecture documentation: met.** The overview page, the generated data dictionary and the diagram set under documentation/ in urpx-org/urpx.
- **Roadmap for the next year: in progress.** Working-group agendas name the candidates (highly dynamic prices at v0.6.0, emissions, an API); no roadmap document in the public repository yet.
- **Growth assessment from LFX Insights: not met.** The project is not onboarded to LFX Insights; git figures are on the Contributions slide.
- **Plan for a thriving user community: in progress.** Public release follows the IP review; co-chair, secretary and outreach roles opened at meeting #018; LF Energy blog post drafted.
- **Outline toward Early Adoption: not met.** Not written yet.

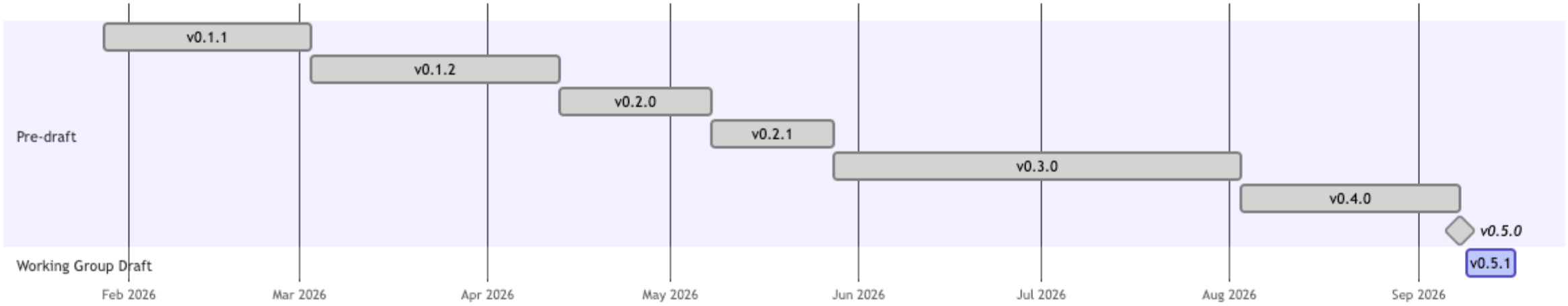
Contributions

urpx-org/urpx, the standard	
Commits by people	257
Commits by the generated-content automation	64
Contributors (people)	4
Organizations contributing commits	Flux Tailor, plus one independent contributor
Maintainers with merge rights	3
Releases tagged	8 (v0.1.1 to v0.5.1)
urpx-org/urpx-wg, the working group	
Commits	47
Contributors (people)	3
Working-group meetings held	19



Figures from git on 2026-10-05. The project is not yet onboarded to LFX Insights.

URPX Releases



Organizations contributing and/or using in production

Contributing to the standard

Flux Tailor

Primary sponsor; chair + 2 contributors

National Laboratory of the Rockies

Collaboration partner

Supporting

Green Button Alliance

Supporter; logo used with permission

UtilityAPI

LF Energy supporter; WG participant

Using in production

- URPX is a Working Group Draft; the repository goes public after the LF Energy IP review.
- URPX serves as foundation for UtiliPub,™ Flux Tailor's rate plan publication platform.
- Further participating organizations are named in the public meeting records of urpx-org/urpx-wg.

Key Achievements in the past year

- Kickoff 2025-11-19; 19 working-group meetings, biweekly since 2026-01-14, open to anyone on the mailing list.
- Seven releases: v0.1.1 (2026-03-03), v0.1.2, v0.2.0, v0.2.1, v0.3.0 (2026-08-03), v0.4.0 and v0.5.0 (2026-09-08); v0.5.1 in preparation.
- Working Group Draft acceptance on 2026-07-17 (LFESS Sandbox to Draft).
- License transition completed 2026-09-09: W3C Document License for the specification, CDLA-Permissive-2.0 for the datasets, Apache-2.0 for code; DCO for contributions.
- Vocabulary namespace at https://urpx.org/ns/ontology/ with a published JSON-LD context; four test-case packages (twelve data files) validating against the SHACL shapes.
- Repository and website publication framework for Working Group organization: urpx (the standard), urpx-dev (development record), urpx-data-sandbox (community data); governance, maintainers, security policy and code of conduct published.
- Brand guidelines and logo package published; LF Energy project page live; Green Button Alliance support.

Repository layout

urpx-org/urpx

The standard. Ontology (OWL), SHACL shapes, the JSON-LD context, SKOS mappings, documentation, test cases, and tooling.

urpx-org/urpx-wg

Meeting agendas and minutes, the brand package, and announcements.

urpx-org/urpx-dev

Development record. Tasks, specifications, working reports, roadmaps, review materials, and worked sample instances.

urpx-org/urpx-data-sandbox

Community data. Contributed datasets for implementation and validation.

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From ontology to published namespace

Author

Ontology, SHACL shapes and the JSON-LD context, authored in urpx.

Generate

GitHub Actions regenerate the data dictionary, the diagrams, the context and the test-case manifests from the ontology and shapes.

Weave

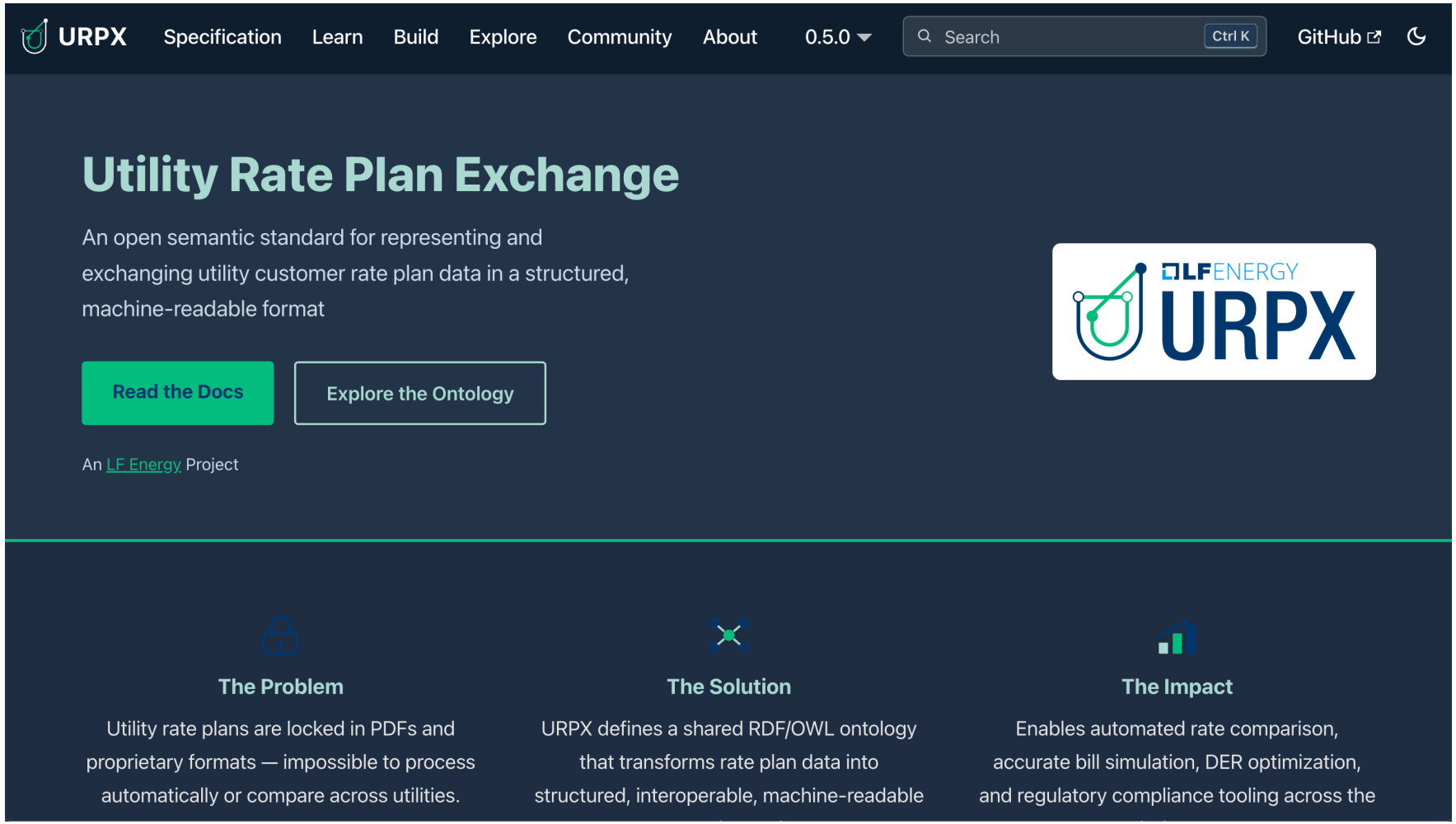
Semantic Flow publishes the ontology, shapes and context as a mesh: one stable page per term, plus immutable release-tagged snapshots.

Publish

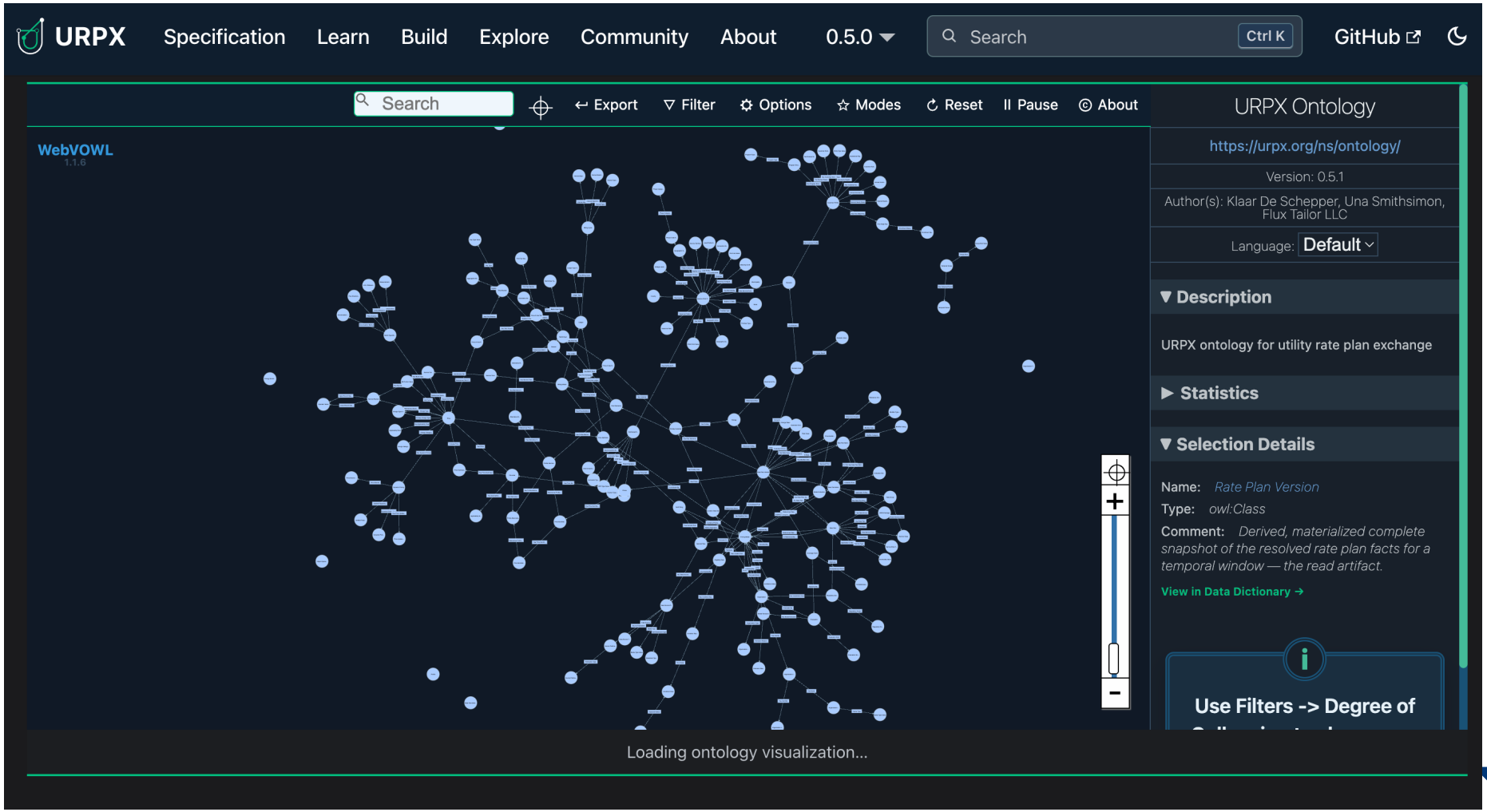
urpx.org/ns/ serves the ontology, the SHACL shapes and the context; the documentation site publishes alongside.

Semantic Flow (weave): github.com/semantic-flow/weave • Namespaces: urpx.org/ns/ontology/, urpx.org/ns/shacl/core/, urpx.org/ns/context

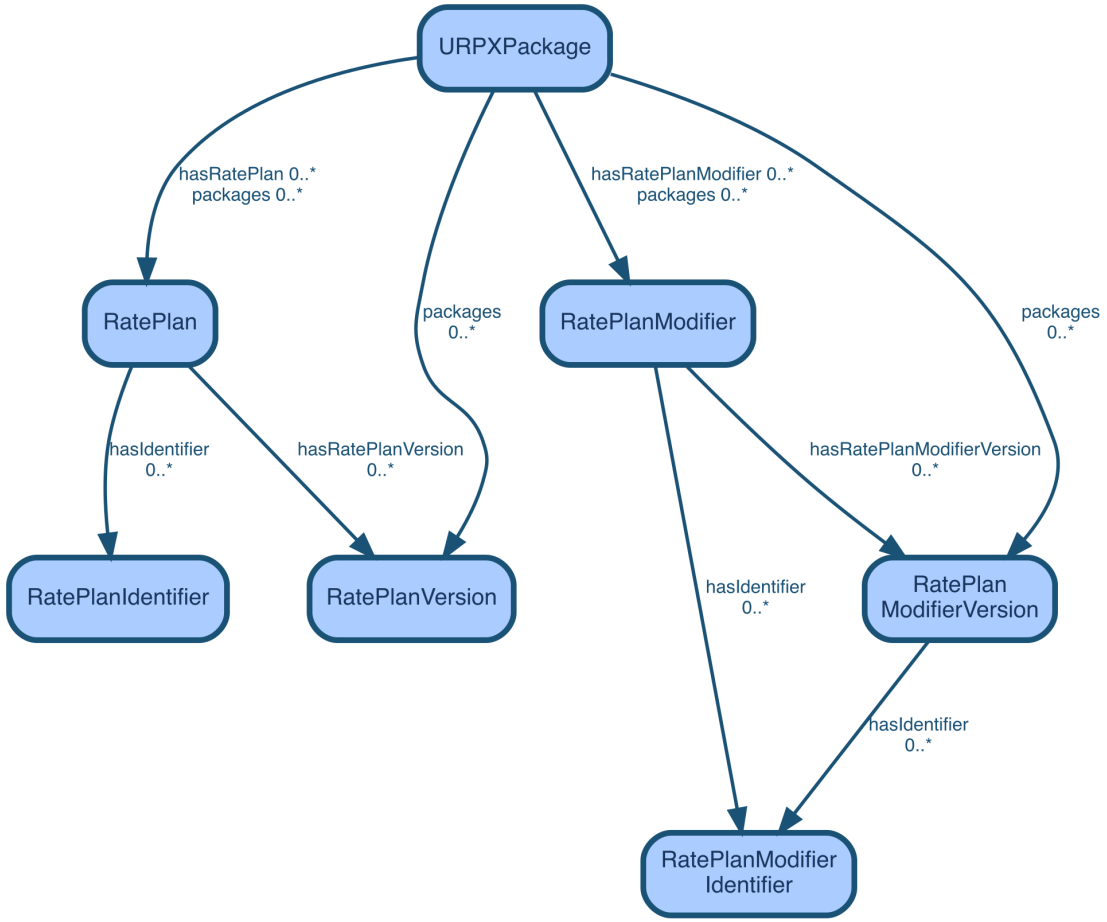
URPX.org website



Documentation: Interactive Ontology Navigator



The model at a glance: core rate plan entities



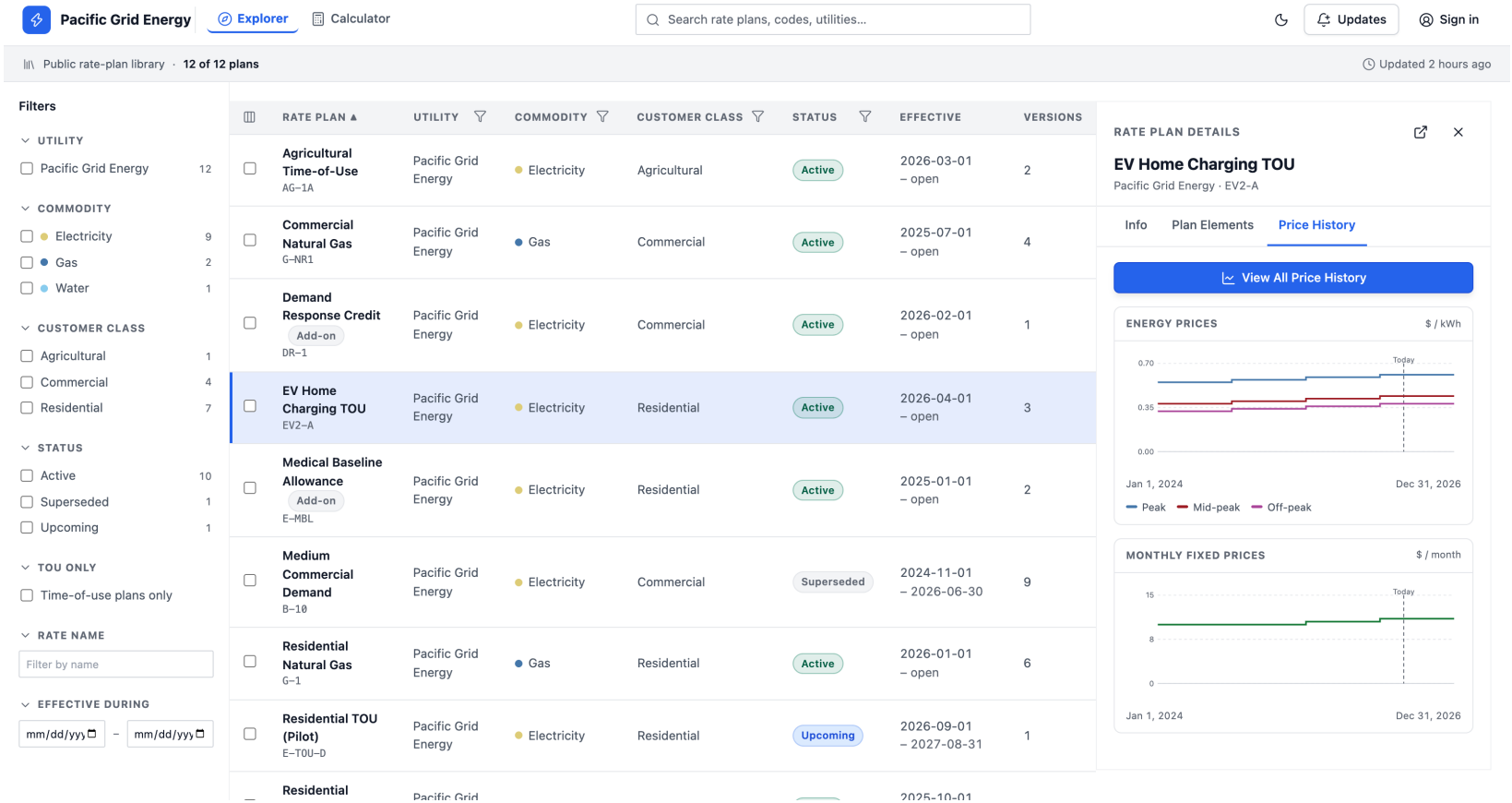
The core rate plan entities and their relationships, generated from the ontology.

Coordinating with grid and market standards

- **Standards alignment:** formal SKOS mappings from URPX to Green Button / NAESB ESPI, LF Energy CDS, CIM and OpenADR is in progress, and interoperability with IEEE 2030.5 (SEP) and Matter. URPX coordinates with usage-data, grid and demand-response standards rather than competing with them.
- **Complementary to CDS:** CDS Customer Data and CDS Registration handle customer-data access and utility discovery; URPX defines the rate plan format those customers are charged under.

URPX in production

- **Reference implementation:** UtiliPub, by Flux Tailor, is a rate plan publication platform built on URPX.



Areas the project could use help on

- **Contributors and committers:** committers from more organizations, and implementers to validate real utility data against the shapes.
  - **API specification:** volunteers to help shape the common service API and a reference implementation.
  - **Working-group roles:** the open co-chair and secretary roles.
  - **CI/CD set up:** to ensure consistency and code quality
  - **Collaborative AI-assisted workflow/harness system**
  - **Launch amplification:** the LF Energy blog post and newsletter at the public release (drafts ready).

Feedback on working with LF Energy

Infrastructure in place: the GitHub organization, the urpx-wg and urpx-committee mailing lists, the LFX Zoom calendar with recordings, the artwork and landscape entries, the project page, and the brand package are all working.

LFX: a lot of surface to learn; staff support has been strong throughout.

Namespace hosting: every URPX term needs a stable, dereferenceable identifier. One harmonized hosting solution across LF Energy projects would let every standards project do this well. URPX solved it with Semantic Flow (github.com/semantic-flow/weave): one page per term, plus immutable release-tagged snapshots.

Standardization across and collaboration between standards